CP2.6 — Hyperbolic Functions

1. Definitions

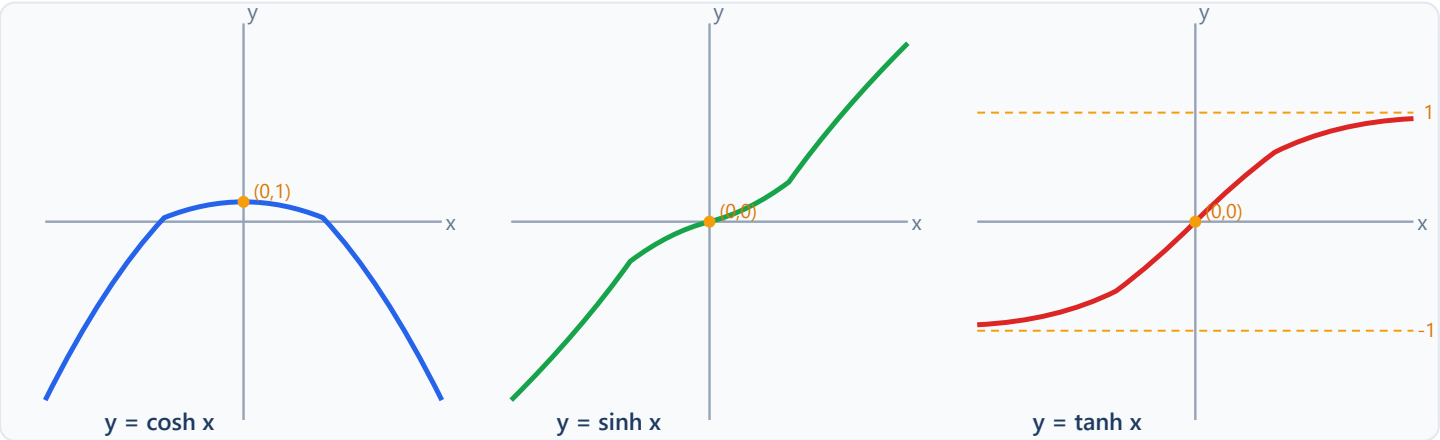
$$\cosh x = \frac{e^x + e^{-x}}{2}, \qquad \sinh x = \frac{e^x - e^{-x}}{2}, \qquad \tanh x = \frac{\sinh x}{\cosh x} = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

Also: $\operatorname{sech} x = \dfrac{1}{\cosh x}$, $\operatorname{cosech} x = \dfrac{1}{\sinh x}$, $\coth x = \dfrac{\cosh x}{\sinh x}$

1.1 Key Values

x	$\cosh x$	$\sinh x$	$\tanh x$
0	1	0	0
$\pm\infty$	$+\infty$	$\pm\infty$	± 1

2. Graphs of Hyperbolic Functions



3. Hyperbolic Identities

Fundamental identity:

$$\cosh^2 x - \sinh^2 x = 1$$

Proof: $\cosh^2 x - \sinh^2 x = \dfrac{(e^x + e^{-x})^2 - (e^x - e^{-x})^2}{4} = \dfrac{4}{4} = 1$

Trig identity	Hyperbolic equivalent
$\cos^2\theta + \sin^2\theta = 1$	$\cosh^2 x - \sinh^2 x = 1$
$\sin 2\theta = 2\sin\theta\cos\theta$	$\sinh 2x = 2\sinh x\cosh x$
$\cos 2\theta = \cos^2\theta - \sin^2\theta$	$\cosh 2x = \cosh^2 x + \sinh^2 x$
$\cos 2\theta = 2\cos^2\theta - 1$	$\cosh 2x = 2\cosh^2 x - 1$
$\cos 2\theta = 1 - 2\sin^2\theta$	$\cosh 2x = 1 + 2\sinh^2 x$
$1 + \tan^2\theta = \sec^2\theta$	$1 - \tanh^2 x = \operatorname{sech}^2 x$
$\cot^2\theta + 1 = \csc^2\theta$	$\coth^2 x - 1 = \operatorname{cosech}^2 x$

Osborne's Rule: To convert a trig identity to a hyperbolic identity, replace:

- $\cos\theta \to \cosh x$
- $\sin\theta \to i\sinh x$ (i.e., replace $\sin^2$ with $-\sinh^2$)
- Every product of two sines (or $\sin^2$) changes sign

This works for identities but does NOT work for inequalities or equations.

Exam tip: Osborne's rule is a memory aid, not a proof. In proofs, start from definitions (exponential form) and manipulate algebraically.

4. Derivatives of Hyperbolic Functions

Function	Derivative
$\sinh x$	$\cosh x$
$\cosh x$	$\sinh x$
$\tanh x$	$\operatorname{sech}^2 x$
$\operatorname{sech} x$	$-\operatorname{sech} x \tanh x$
$\operatorname{cosech} x$	$-\operatorname{cosech} x \coth x$
$\coth x$	$-\operatorname{cosech}^2 x$

Proof for tanh: $\dfrac{d}{dx}(\tanh x) = \dfrac{\cosh x \cosh x - \sinh x \sinh x}{\cosh^2 x} = \dfrac{1}{\cosh^2 x} = \operatorname{sech}^2 x$

5. Integration of Hyperbolic Functions

Integral	Result
$\int \sinh x \, dx$	$\cosh x + C$
$\int \cosh x \, dx$	$\sinh x + C$
$\int \tanh x \, dx$	$\ln(\cosh x) + C$
$\int \operatorname{sech}^2 x \, dx$	$\tanh x + C$
$\int \sinh^2 x \, dx$	$\frac{1}{4}\sinh 2x - \frac{x}{2} + C$
$\int \cosh^2 x \, dx$	$\frac{1}{4}\sinh 2x + \frac{x}{2} + C$

6. Inverse Hyperbolic Functions

$$\operatorname{arsinh} x = \ln\left(x + \sqrt{x^2+1}\right), \quad x \in \mathbb{R}$$

$$\operatorname{arcosh} x = \ln\left(x + \sqrt{x^2-1}\right), \quad x \geq 1$$

$$\operatorname{artanh} x = \frac{1}{2}\ln\left(\frac{1+x}{1-x}\right), \quad |x| < 1$$

6.1 Proof for arsinh x

Let $y = \operatorname{arsinh} x$, so $\sinh y = x$, i.e. $\dfrac{e^y - e^{-y}}{2} = x$.

Multiply by $2e^y$: $e^{2y} - 1 = 2xe^y$, so $e^{2y} - 2xe^y - 1 = 0$.

Quadratic in e^y: $e^y = \dfrac{2x \pm \sqrt{4x^2+4}}{2} = x \pm \sqrt{x^2+1}$.

Since $e^y > 0$ and $x - \sqrt{x^2+1} < 0$, we take the positive root: $e^y = x + \sqrt{x^2+1}$, so $y = \ln(x + \sqrt{x^2+1})$. ✓

7. Derivatives of Inverse Hyperbolic Functions

$$\frac{d}{dx}(\operatorname{arsinh} x) = \frac{1}{\sqrt{x^2+1}}$$

$$\frac{d}{dx}(\operatorname{arcosh} x) = \frac{1}{\sqrt{x^2-1}}, \quad x > 1$$

$$\frac{d}{dx}(\operatorname{artanh} x) = \frac{1}{1-x^2}, \quad |x| < 1$$

Proof for arsinh: Let $y = \operatorname{arsinh} x$, so $\sinh y = x$. Differentiate implicitly: $\cosh y \cdot \dfrac{dy}{dx} = 1$. So $\dfrac{dy}{dx} = \dfrac{1}{\cosh y} = \dfrac{1}{\sqrt{\sinh^2 y + 1}} = \dfrac{1}{\sqrt{x^2+1}}$.

7.1 General Forms (Chain Rule)

Function	Derivative
$\operatorname{arsinh}\left(\frac{x}{a}\right)$	$\frac{1}{\sqrt{x^2+a^2}}$
$\operatorname{arcosh}\left(\frac{x}{a}\right)$	$\frac{1}{\sqrt{x^2-a^2}}$

8. Integration Using Inverse Hyperbolic Functions

$$\int \frac{1}{\sqrt{x^2+a^2}}\,dx = \operatorname{arsinh}\left(\frac{x}{a}\right) + C = \ln\left(x+\sqrt{x^2+a^2}\right) + C$$

$$\int \frac{1}{\sqrt{x^2-a^2}}\,dx = \operatorname{arcosh}\left(\frac{x}{a}\right) + C = \ln\left(x+\sqrt{x^2-a^2}\right) + C, \quad x > a > 0$$

Example: Find $\int \frac{1}{\sqrt{x^2+9}}\,dx$.

$a = 3$. $\operatorname{arsinh}\left(\frac{x}{3}\right) + C = \ln\left(x+\sqrt{x^2+9}\right) + C$.

Example: Find $\int \frac{dx}{\sqrt{x^2+4x+8}}$.

Complete the square: $x^2+4x+8 = (x+2)^2+4$.
Let $u = x+2$: $\int \frac{du}{\sqrt{u^2+4}} = \operatorname{arsinh}\left(\frac{u}{2}\right) + C = \operatorname{arsinh}\left(\frac{x+2}{2}\right) + C$.

Example: Evaluate $\int_2^3 \frac{1}{\sqrt{x^2-1}}\,dx$.

$a = 1$. $[\operatorname{arcosh} x]_2^3 = \operatorname{arcosh} 3 - \operatorname{arcosh} 2 = \ln(3+\sqrt{8}) - \ln(2+\sqrt{3})$.
$= \ln(3+2\sqrt{2}) - \ln(2+\sqrt{3})$.

Exam tip: When integrating with inverse hyperbolics, you can write either the arsinh form or the $\ln$ form — both are accepted. The ln form is sometimes required if the question says "in the form $\ln(\ldots)$".

9. Hyperbolic Substitutions for Integration

Integrand type	Substitution	Reason
$\sqrt{a^2+x^2}$	$x = a\sinh t$	$\cosh^2 t - \sinh^2 t = 1 \Rightarrow \sqrt{a^2 + a^2\sinh^2 t} = a\cosh t$
$\sqrt{x^2-a^2}$	$x = a\cosh t$	$\sqrt{a^2\cosh^2 t - a^2} = a\sinh t$

Example: Find $\int \sqrt{x^2+4}\,dx$ using the substitution $x = 2\sinh t$.

$dx = 2\cosh t\,dt$, $\sqrt{x^2+4} = \sqrt{4\sinh^2 t + 4} = 2\cosh t$.

$$\int 2\cosh t \cdot 2\cosh t\,dt = 4\int \cosh^2 t\,dt = 4\int \frac{1+\cosh 2t}{2}\,dt = 2t + \sinh 2t + C$$

$$= 2t + 2\sinh t\cosh t + C$$

Back-substitute: $t = \operatorname{arsinh}(x/2)$, $\sinh t = x/2$, $\cosh t = \frac{\sqrt{x^2+4}}{2}$:

$$= 2\operatorname{arsinh}\left(\frac{x}{2}\right) + \frac{x\sqrt{x^2+4}}{2} + C$$

10. Summary

Identity	Formula
Fundamental	$\cosh^2 x - \sinh^2 x = 1$
Double angle	$\cosh 2x = \cosh^2 x + \sinh^2 x = 2\cosh^2 x - 1 = 1 + 2\sinh^2 x$

$d/dx\ \sinh$	$\cosh x$
$d/dx\ \cosh$	$\sinh x$ (note: not negative!)
$d/dx\ \tanh$	$\operatorname{sech}^2 x$
$d/dx\ \operatorname{arsinh} x$	$1/\sqrt{x^2+1}$
$d/dx\ \operatorname{arcosh} x$	$1/\sqrt{x^2-1}$
$\int 1/\sqrt{x^2+a^2}$	$\operatorname{arsinh}(x/a)+C$
$\int 1/\sqrt{x^2-a^2}$	$\operatorname{arcosh}(x/a)+C$

11. Exam-Style Practice Questions

1. **[Proof]** Starting from the definitions of $\sinh x$ and $\cosh x$, prove that $\cosh 2x = 2\cosh^2 x - 1$.
2. **[Proof]** Show that $\operatorname{arcosh} x = \ln(x + \sqrt{x^2-1})$ for $x \geq 1$.
3. **[Differentiation]** Differentiate (a) $y = \sinh(x^2)$, (b) $y = x\cosh 3x$, (c) $y = \operatorname{arsinh}(2x+1)$.
4. **[Integration]** Find $\displaystyle\int_0^2 \cosh^2 x\, dx$.
5. **[Completing the square]** Find $\displaystyle\int \frac{dx}{\sqrt{2x^2+4x+6}}$.
6. **[Solving equations]** Solve $2\cosh x - \sinh x = 2$ giving your answer in logarithmic form.
7. **[Identity]** Prove that $\tanh^2 x + \operatorname{sech}^2 x = 1$.

General exam tips for CP2.6:

- Key difference from circular trig: $\dfrac{d}{dx}\cosh x = +\sinh x$ (not negative!)
- arsinh is defined for all $x \in \mathbb{R}$; arcosh is defined only for $x \geq 1$
- For "solve $a\cosh x + b\sinh x = c$" — substitute $e^x = u$ and solve the resulting quadratic
- For inverse hyperbolic proofs: set $y = \operatorname{arsinh} x$ (or arcosh), get quadratic in e^y, take the valid root